SAEINDIA™
SOUTHERN SECTION

Autonomous Drone Development Challenge (ADDC) 2026

RULE BOOK

SAE INDIA Southern Section
Visit: https://saeiss.org/ Email: addc@saeiss.org

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Foreword

Dear Participants, It gives us immense pleasure to present to you the rule book for SAE Autonomous Drone Development Challenge 2024. We would like to extend our warmest welcome to a you who are participating in this competition. The rule book has been designed to provide you with al necessary information regarding the competition. It contains detailed guidelines for the competi including the mission requirements, scoring calculation, and the two parts of the competition - Static Dynamic Events. We believe that this competition will not only challenge your technical skills but enhance your interpersonal and communication skills. The competition aims to solve real-world prob with multirotor aerial vehicles, and we hope that your participation will contribute towards achieving objective. We urge you to thoroughly read and understand the rules and regulations mentioned in this b to ensure a safe and fair competition for everyone. We also encourage you to demonstrate your best s and creativity and wish you all the best for the competition. Let's make this a memorable event and may best team win!

Best regards,

SAE ISS Autonomous Drone Development Rules Committee Members, Aerosp Development Council

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SECTION 1
INTRODUCTION

Drone research is closely related to the growing enthusiasm for robotics as a cheaper alternative to high-priced manned piloted systems. Remotely Piloted Aircraft Systems (RPAS) have shrunk and become more affordable because of recent advancements in processing, battery, and sensor device technologies. In particular, affordable, easily accessible micro-electromechanical sensors have helped to advance the development of RPAS autopilots in a number of fields. Successful applications include law enforcement, academic research, military and non-military use, and limited recreation. Compared to manned air vehicles, they can be built, produced, and operated for a much lower price. The actual reductions in power, air platform, and engine usage.

The SAEISS Autonomous Drone Development Challenge aims to advance Indian university and college-level knowledge and experience in unmanned systems technologies. Even tiny unmanned vehicles are intricate systems that require careful planning and execution during the design phase. In addition, just like in any other vehicle design project, safety considerations are crucial in this competition. The goal of the competition is to present a practical engineering challenge to engineering students at the undergraduate and graduate levels. It has been created to expose students to the kinds of scenarios that engineers deal with in their everyday workplaces. Each team must come up with, design, and create a multirotor UAV prototype that satisfies the mission criteria

There are two parts to each competition, each with their own series of events. These parts are known as Static and Dynamic Events. The events award the team points towards a grand total. The team with the greatest total wins the competition overall. Some events, such as tech inspection, award no points but are required in order to proceed to other events for safety reasons. Teams accomplishing the assigned mission fully are only eligible for the grand prize.

SECTION 2

COMPETITION GENERAL REQUIREMENTS

2.1 OFFICIAL ANNOUNCEMENTS AND COMPETITION INFORMATION

2.1.1 SAEISS Autonomous Drone Development Challenge

Use of drones is becoming popular among the public and the technology involves human pilots to control their aerial vehicles to fly at different speeds through complex environments and pass a number of hurdles in the real world. A system including navigation and control for autonomous drone operations is highly demanding. Drone logistics is a potential segment where investors and multinational companies are pouring in investments and medical delivery drones are a great point of interest.

2.1.2 SAE Aero Design Rules and Organizer Authority Rules

The SAE ISS Autonomous Drone Development Challenge Rules are the responsibility of the SAE ISS Autonomous Drone Development Competition Committee and are issued under the authority of the SAEISS Aerospace Development Council. Official announcements from the SAEISS Autonomous Drone Development Challenge Committee, SAEISS Aerospace Development Council shall be considered part of and have the same validity as these rules.

Ambiguities or questions concerning the meaning or intent of these rules will be resolved by the officials, SAEISS Autonomous Drone Development Challenge Rules Committee or SAEISS Staff.

2.1.3 Rules Validity

The SAE ISS Autonomous Drone Development Challenge Rules posted on the SAEISS Website and dated for the calendar year of the competition are the rules in effect for the competition. Rule sets dated for other years are invalid.

2.1.4 Understanding the Rules

Teams are responsible for reading and understanding the rules in its entirety effect for the competition in which they are participating. The section and paragraph headings in these rules are provided to facilitate reading: they do not affect the paragraph contents.

2.1.5 Loopholes

It is virtually impossible for a set of rules to be so comprehensive that it covers all possible questions about the plane's design parameters or the conduct of the competition.

Please keep in mind that safety remains paramount during any SAEISS competition, so any perceived loopholes should be resolved in the direction of increased safety/ concept of the competition.

2.1.6 Participating in the Competition

Teams, team members as individuals, faculty advisors and other representatives of a registered university who are present on-site at a competition are considered to be "Participating in the competition" from the time they arrive at the event site until they depart the site at the conclusion of the competition or earlier by withdrawing.

2.1.7 Violations of Intent

The violations of the intent of a rule will be considered a violation of the rule itself. Questions about the intent or meaning of a rule may be addressed to the SAEISS Officials, Competition Organizers or SAEISS Staff.

2.1.8 Right to Impound

SAEISS and the other competition organizing bodies reserve the right to impound any on-site aerial vehicle at any time during a competition for inspection and examination by the organizers, officials and technical inspectors.

2.1.9 General Authority

SAEISS and the competition organizing bodies reserve the right to revise the schedule of any competition and/or interpret or modify the competition rules at any time and in any manner that is, in their sole judgment, required for the efficient operation of the event or the SAEISS Autonomous Drone Development Challenge series as a whole.

2.1.10 Penalties

Organizers have the right to modify the points and/or penalties listed in the various event descriptions; to better reflect the design of their events, or any special conditions unique to the site.

2.2 TEAM MEMBER ELIGIBILITY

Teams are required to read the articles posted on the SAEISS Website published by SA Aerospace Development. Teams must also be familiar with all official announcements concernin competitions and rule interpretations released by the SAEISS Autonomous Drone Develop Competition Rules Committee.

2.3 SOCIETY MEMBERSHIP

Individual team members must be members of SAE India. Proof of membership, such membership card, is required at the event.

For membership details please visit www.saeindia.org

2.3.1 Liability Waiver and Insurance Requirements

All on-site participants and faculty advisors are required to sign a liability waiver upon registrati Individual medical and accident insurance coverage is the sole responsibility of the participant.

2.4 RINGERS PROHIBITED

In order to maintain the integrity of a fair competition, the faculty advisor must prohibit ringers. ringer is someone that has exceptional skills related to the competition (e.g., a professional model build that cannot be a legal member of the team but helps the team win points.

2.5 DESIGN AND FABRICATION

The aerial vehicle shall be conceived, designed, integrated, fabricated, and tested primarily by SA INDIA student members of the participating team. Teams are encouraged to seek technical guidance an mentorship from faculty members, industry experts, and professionals to enhance their understanding engineering concepts, design methodologies, and best practices. Such guidance should be advisory in natur and should support the students' learning process. The final engineering decisions, system integration design documentation, fabrication, software development, testing, and overall implementation shall remai the responsibility of the student team. All significant technical guidance, external references, software tools and supporting resources utilized during the project shall be appropriately acknowledged in the Technica Design Report.

The faculty advisor shall certify that the project represents the original work of the student team by signing the Statement of Compliance provided in the Appendix.

2.6 OFFICIAL LANGUAGES

The official language of the SAEISS Autonomous Drone Development Challenge series is English. Document submissions, presentations and discussions in English are acceptable at all competitions in the series.

2.7 REGISTRATION INFORMATION, DEADLINES AND WAITLIST

Teams intending to participate in the SAEISS Autonomous Drone Development Challenge must register online as per the schedule mentioned in the official website. The registration fee is non-refundable and failure to meet these deadlines will be considered a failure to qualify for the competition.

Individual Registration Requirements – ACTION REQUIRED

If you are not an SAE India member, go to www.saeindia.org and select the "Membership" link. Students will need to select the "Student Membership" link and then follow the series of questions that are asked Please note all student participants must be SAE India members to participate in the events. Faculty members who wish to become SAE India members should choose the "Professional Membership" link.

2.8 POLICY DEADLINE (FAILURE TO MEET DEADLINES)

Teams registering for SAEISS Autonomous Drone Development Challenge are required to submit a design report prior to the competition. The event judges need to evaluate the team during the competition based on the reports submitted. When the document is not submitted our judges cannot properly assess the team. Additionally, teams that do not submit a Design Report typically do not come to the competition. Teams that do not notify us that they are withdrawing create the following problems

(1) They are included in the static event schedules and judging time is wasted.

(2) Their unused registration slot cannot be offered to a team on the waitlist. Additionally, failure to submit the required Design Report is a clear violation of the rules (Need ruling where you cannot fly unless competed design submission)

Therefore SAEISS will be placing the policy into effect, that failure to submit the required De Report within 10 days of the deadline will constitute an automatic withdrawal of your team. The team be notified after the 9th day of no submission that we have not received your documents and after th days your team's registration will be cancelled and no refund will be given.

2.9 FACULTY ADVISOR

Each team is expected to have a Faculty Advisor appointed by the university. The Faculty Adv is expected to accompany the team for the competition and will be considered by competition officia be the official university representative. Faculty Advisors may advise their teams on general enginee and engineering project management theory. Faculty advisors with SAE India Professional member accompanying the teams to the finals will have points added to their team's total points.

2.10 QUESTIONS, COMPLAINTS AND APPEALS

2.10.1 Questions

Any questions or comments about the rules should be brought to the attention of the Ru Committee via the SAEISS Autonomous Drone Development Challenge email: addc@saeiss.org

2.10.2 Complaints

Competition officials will be available to listen to complaints regarding errors in scorin interpretation, or application of the rules during the competition. Competition officials will not be availab to listen to complaints regarding the nature, validity, or efficacy of the rules themselves at the competitio In other words, the Organizer will not change the rulebook at the field.

2.10.3 Appeal / Preliminary Review

A team can only appeal issues related to own-team scoring, judging, venue policies, and/or an official actions. Team Captain(s) and/or faculty advisor must bring the issue to the Organizer's or SAEIS staff's attention for an informal preliminary review before filing an official appeal. A team cannot file a appeal to cause harm to another team's standing and/or score.

2.10.4 Cause for Appeal

A team may appeal any rule interpretation, own-team scoring or official actions) which the team feel has caused some actual, non-trivial, harm to own-team, or has had a substantive effect on their score Teams may not appeal rule interpretations or actions that have not caused them any substantive damage.

2.10.5 Appeal Format

If a faculty advisor or team captain(s) feel that their issue regarding an official action or rules interpretation was not properly addressed by the event officials, the team may file a formal appeal to the action or rules interpretation with the Appeals Committee.

All appeals must be filed in writing to the Organizer by the faculty advisor or team Captain only. All appeals will require the team to post twenty five (25) points as collateral. If the appeal is successful and the action is reversed, the team will not forfeit the twenty five (25) collateral points. If the appeal is overruled, the team will forfeit the twenty five (25) collateral points.

All rulings issued by the Appeals Committee are final.

2.10.6 PROFESSIONAL CONDUCT

Conduct

In the event of unsportsmanlike conduct by team members or that team's faculty advisor, the team will receive a warning from a Competition Official. A second violation will result in expulsion of the team from the competition and loss of any points earned in all aspects of the competition.

Arguments with Officials

Arguments with or disobedience toward any competition official may result in the team being eliminated from the competition. All members of the team may be immediately escorted from the grounds.

Alcohol and Illegal Material

Alcoholic beverages, illegal drugs, firearms, weapons, or illegal material of any type are not permitted on the event sites at any time during the competition. Any violations of this rule will result in the immediate expulsion of all members of the offending school, not just the individual team member in violation. This rule applies to team members and faculty advisors. Any use of illegal drugs or any use of alcohol by an underage person must be reported to the local law enforcement authorities for prosecution.

Organizer's Authority

The Organizer reserves the exclusive right to revise the schedule of the competition and/or to interpret the competition rules at any time and in any manner which is required for efficient operation or safety of the competition.

SECTION 3

Dynamic Mission Rules

3.1.1 Theme

Human-Machine Interaction (HMI)

Mission: Autonomous Reconnaissance and Intelligence Recovery

3.1.2 Introduction

SAE ADDC 2026 challenges teams to develop an autonomous UAV that collaborates with a huma operator to complete a reconnaissance mission within five minutes.

3.1.3 Mission Statement

The UAV autonomously surveys the arena, locates the intelligence cache, scans a QR code an decodes two digits of a four-digit lock code. Simultaneously, the ground operative completes a puzzl course to obtain the remaining two digits. The combined code unlocks the intelligence cache, after whic the team retrieves the intelligence package and returns to the finish zone.

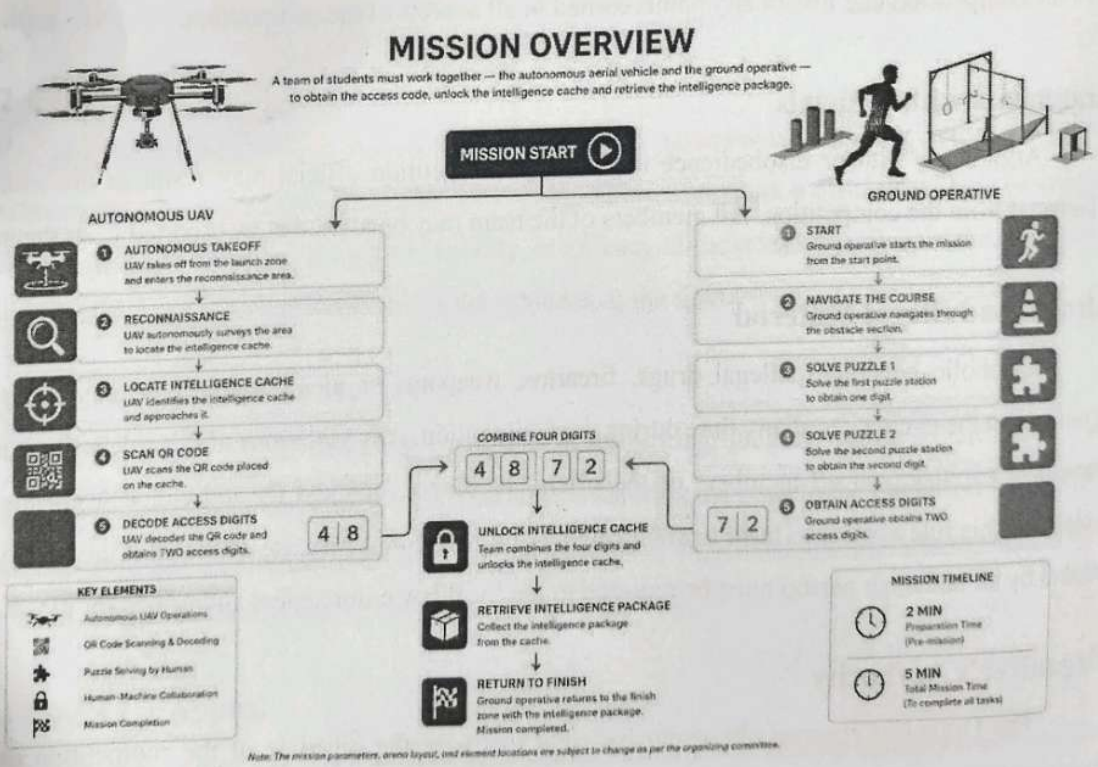


Figure 1.Misssion overview

3.2 Round Attempt

1. Each team shall be permitted one (1) mission attempt per competition round.
2. Teams will be provided with two (2) minutes for mission preparation before the commencement of the dynamic event.
3. During the preparation period, teams may power up the UAV, verify system readiness, and perform pre-flight safety checks. The UAV shall remain disarmed until instructed by the Flight Marshal.
4. At the end of the two-minute preparation period, teams shall immediately initiate the autonomous mission.
5. The complete mission, including autonomous reconnaissance, ground operative tasks, intelligence retrieval, and return to the finish zone, shall be completed within five (5) minutes.
6. The mission timer shall begin immediately after the official start signal issued by the organizing committee.
7. The UAV shall autonomously survey the designated reconnaissance area, locate the intelligence cache, detect the QR code mounted on the cache, and decode the two access digits.
8. Simultaneously, the designated ground operative shall complete the prescribed obstacle and puzzle course to obtain the remaining two access digits.
9. The intelligence cache may only be unlocked after the team has successfully combined all four access digits.
10. Mission completion is achieved only when the intelligence package has been retrieved and the ground operative returns to the designated finish zone within the stipulated mission time.

3.3 Competition Arena Requirements

1. During the mission, the UAV shall operate autonomously without manual flight control.
2. Manual intervention shall be permitted only in the event of an emergency or if instructed by the Flight Marshal for safety reasons.
3. Any manual takeover of the UAV shall result in the mission being considered aborted and no dynamic mission score shall be awarded unless otherwise specified by the organizing committee.
4. The UAV shall remain within the designated flight boundaries throughout the mission.
5. The ground operative shall remain within the designated obstacle and puzzle course.
6. Teams shall not enter restricted areas or interfere with other competing teams.

7. The mission shall be considered complete only when:
   - ✓ the UAV has successfully completed the reconnaissance task,
   - ✓ the intelligence cache has been unlocked,
   - ✓ the intelligence package has been retrieved, and
   - ✓ the ground operative has returned to the designated finish zone.

3.4 Launch

1. The UAV launch location shall be assigned by the Organizing Committee.
2. The position of the intelligence cache, obstacle course, and puzzle stations shall be deter by the Organizing Committee before each mission.
3. Teams shall not enter the reconnaissance area before the official mission start.
4. The mission shall commence only after the official start signal from the Flight Marshal.

3.5 Landing

1. A successful landing is defined as a controlled autonomous landing within the designated l zone.
2. The landing shall occur without causing damage to the UAV or creating unsafe condition
3. Deviations from the designated landing area may result in score deductions as specified i scoring criteria.
4. If the mission is completed before landing, the UAV shall return to the designated h location and perform a safe landing to conclude the flight.

3.6 Intelligence Cache and QR Code Mission

1. A single QR code shall be mounted on the intelligence cache.
2. The UAV shall autonomously detect and scan the QR code.
3. The QR code shall reveal two digits of the four-digit intelligence access code.
4. The remaining two digits shall be obtained by the ground operative through succes completion of the designated puzzle stations.
5. The intelligence cache may only be unlocked after all four digits have been corre identified and combined.
6. The organizing committee may vary the location of the intelligence cache and QR code different mission attempts.

7. Teams are responsible for ensuring successful QR code recognition before attempting to unlock the intelligence cache.
8. The intelligence package shall be retrieved only after the cache has been successfully unlocked.

3.7 GROUNDING OF AN AIRCRAFT

An aircraft may be grounded if it is determined by any SAE India Southern Section official, Technical Inspector, Safety Officer, or Event Official to be unsafe, non-flight-worthy, or non-compliant with the competition regulations. The aircraft shall not be permitted to participate further until the identified issues have been rectified and the UAV has successfully passed re-inspection.

3.8 NO-FLY ZONES

This section can remain almost identical, replacing "payload mission" references with "reconnaissance mission."

3.9 FLIGHT BRIEFING

Prior to the commencement of the dynamic event, all teams shall attend the mandatory Flight and Mission Briefing conducted by the Organizing Committee. The briefing shall include flight boundaries, no-fly zones, mission procedures, safety regulations, emergency protocols, and any arena-specific instructions.

3.10 RULE VIOLATIONS

Violation of any safety, flight, or mission rule may result in penalties, invalidation of the mission attempt, or disqualification at the discretion of the Organizing Committee. The decision of the Chief Judge and Event Officials shall be final.

Dynamic Mission Points	
Mission Task	**Points**
Autonomous Take-off	20
Reconnaissance & Navigation	40
Intelligence Cache Detection	30
QR Code Detection & Decoding (2 Digits)	40
Ground Puzzle Completion (2 Digits)	40
Unlocking Intelligence Cache	30
Retrieval of Intelligence Package	20
Return to Finish Zone	20
Safe Mission Completion	10
Total	**250**

SECTION 4

The event will be divided into 3 phases (PHASE-1 & PHASE-2 ARE CONSIDERED AS VIRTUAL CHALLENGE) as follows:

Phase 1: Design Report: Teams will electronically submit their proposals for competition detailing how their design has met or exceeded the design requirements.

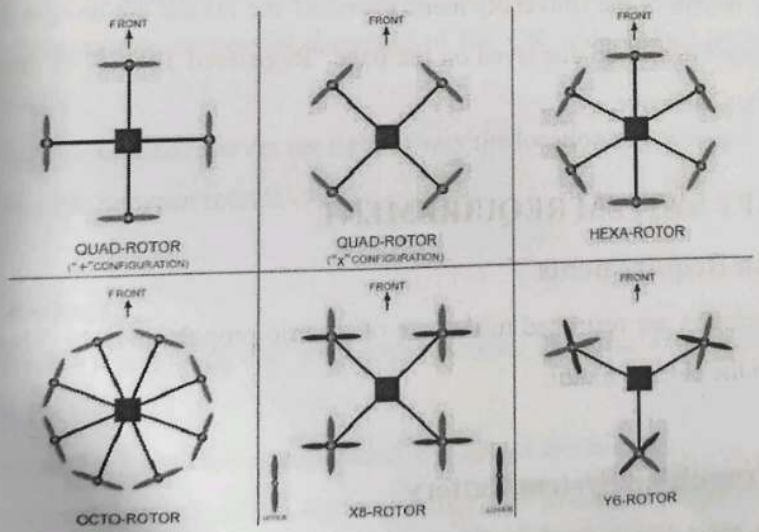
Phase 2: Technical Inspection and Flight Round: Technical inspection will be carried out before the flight round.

Phase 3: Technical Presentation

Phase 3 – Oral Presentation. The oral presentation may be conducted after Flight rounds.

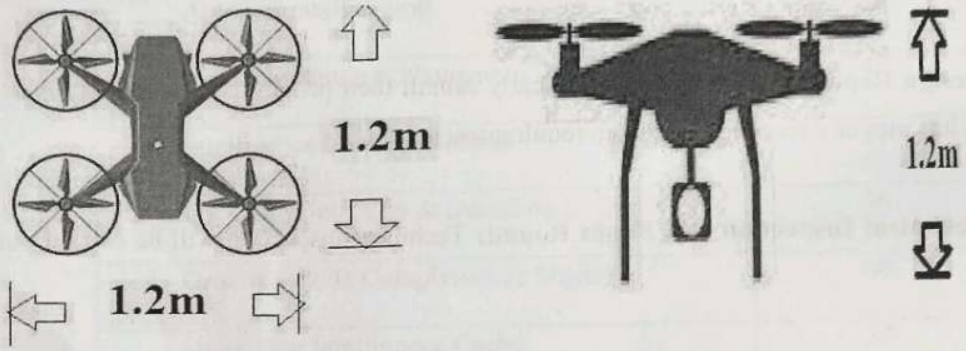
4.1 Aircraft Configuration

Competing designs can use any multirotor configuration. No lighter-than-air flying vehicles are only restricted



4.2 AIRCRAFT DIMENSION REQUIREMENT

Fully configured for takeoff, the free standing aircraft shall have a maximum length, width, an height of 1.2m. Aircraft exceeding this design requirement will be disqualified from the competition.



4.2.1 Aircraft Weight Limit (Including Payload)

The total weight of the aircraft must not be more than 2kg.

4.2.2 Aircraft Identification

Team number as assigned by SAEISS must be visible on the model. The University/College initial may be substituted instead of the University name provided the initials are unique and recognizable. Th assigned aircraft numbers will be displayed on the page "Registered Teams" of http://saeiss.org/studen members/aero-design-challenge.

4.3 AIRCRAFT SYSTEM REQUIREMENT

4.3.1 Propulsion Requirements

All aerial vehicles are restricted to the use of electric propulsion only. There are no restriction (make or model) on the electric motor.

4.3.2. Aircraft Propulsion System Battery

The models must be powered by a commercially available Lithium-polymer battery pack an homemade batteries are NOT allowed.

4.4 Intelligence Cache Requirements

4.4.1 Intelligence Cache and QR Code

- The Intelligence Cache represents the mission objective and shall be placed within the designated reconnaissance area by the Organizing Committee.
- The Intelligence Cache shall contain a four-digit electronic lock or equivalent locking mechanism.
- A single QR code shall be mounted on the top surface of the Intelligence Cache in a clearly visible position.
- The QR code shall contain two (2) digits of the four-digit access code.
- The remaining two (2) digits shall be obtained by the ground operative through successful completion of the designated puzzle stations.
- Teams shall combine all four digits to unlock the Intelligence Cache.
- The Intelligence Cache shall not be moved, tampered with, or opened by any means other than using the correct access code.
- Any attempt to force open, damage, or bypass the locking mechanism shall result in mission failure.

4.4.2 QR Code Detection Requirements

- The UAV shall autonomously detect and scan the QR code during the reconnaissance mission.
- The QR code shall be scanned without physical contact with the Intelligence Cache.
- Teams shall demonstrate successful decoding of the QR code before attempting to unlock the Intelligence Cache.
- The Organizing Committee reserves the right to vary the location and orientation of the Intelligence Cache and QR code between rounds.

4.4.3 Intelligence Package

- Upon successfully unlocking the Intelligence Cache, the ground operative shall retrieve the Intelligence Package.
- The Intelligence Package shall remain intact throughout the mission.
- The retrieved Intelligence Package shall be carried back to the designated Finish Zone.
- The mission shall be considered complete only when the Intelligence Package is returned to the Finish Zone within the specified mission time.

Note

- ***Teams shall not touch or manipulate the Intelligence Cache until the complete four-di access code has been obtained.***
- ***Only the designated ground operative is permitted to retrieve the Intelligence Package aft the cache has been successfully unlocked.***
- ***The UAV shall not make physical contact with the Intelligence Cache or QR code duri the reconnaissance mission.***

4.4 GENERAL REQUIREMENTS

- **Radios: The use of 2.4 GHz radio is required for all aircraft competing.**
- **Propellers: Metal propellers are not allowed.**

Propeller guards: To ensure safety, all aerial must have propeller guards or a creative design to shield propellers that have the possibility of having direct impact with external objects.

4.6 Scoring sheet

Score	
Description	**Maximum points**
Faculty Advisor in Finals	25
Workshop	25
Technical Design report (TDR)	100
Technical Presentation	100
Dynamic Mission (Reconnaissance & HMI)	250
Total points	**500**
Penalty for Late Submission of Design Report	
For One Day	5
Penalty during Oral Presentation	
Oral presentation exceeds 10 minutes	5
Penalty during Technical Inspection	
Miscellaneous	5
Failure to Report Design Changes	5

SECTION 5
DESIGN REPORT

A Design Report is a crucial document that outlines the thought processes and engineering philosophy used by a team to design an aircraft that's most suitable for a specific mission. It covers various topics such as vehicle configuration, wing plan form, drag analysis, stability and control, power plant performance, and performance prediction. The report provides a comprehensive understanding of how the team arrived at their design conclusions and helps stakeholders evaluate the feasibility and effectiveness of the proposed aircraft.

5.1 SUBMISSION DEADLINES

The Technical Design Report, 2D drawing, and supplemental Tech Data Sheet (TDS) must be electronically submitted to **addc@saeiss.org** no later than the deadlines indicated on **http://saeiss.org/student-members/aero-design-challenge/**. Neither the Organizer nor the SAE ISS is responsible for any lost or misdirected reports, drawings, or server routing delays. The SAE ISS will not receive any hard/scanned copies of the reports through regular mail or email.

5.2 2D DRAWINGS

1.2 D Format and Size

The 2D drawing must be A3 sized page (PDF) format (11 x 17 inches).

1. Drawing shall consist of one (1) page
2. Markings Required.

The 2D drawing must be clearly marked with: Team number, Team name and University / College name.

2. Views Required

The 2D drawing must include a standard aeronautical three-view of the aircraft arranged as described below:

1. Left side view in the lower left with the nose pointing left.
2. Top view above the left side view also with the nose pointing left.
3. Front view in the lower right.

3. Dimensions Required

At a minimum, all aircraft drawings must have the following dimensions clearly shown:

1. Aircraft length, width and height.
2. Drawing shall have CG locations clearly dimensioned in reference to the aircraft datum.
3. All drawing dimensions must be in mm, to an appropriate level of precision.

Technical Design report points		
S.No	**Description**	**Points**
1	Mission concept and system architecture	20
2	UAV Design & Integration	20
3	Autonomous navigation strategy	15
4	Computer vision and QR Detection	15
5	Structural / Aerodynamic / any other analysis	10
6	Safety and Risk assessment	10
7	Unique Technological development	10
	Total	**100**

Table 5.1

Grand Prize

The Grand Prize shall be awarded to the team that achieves the highest overall score while successfully completing all mandatory mission objectives in accordance with the competition rules. To be eligible for the Grand Prize, a team shall successfully:

- Complete the autonomous reconnaissance mission.
- Locate the Intelligence Cache.
- Autonomously detect and decode the QR code to obtain the assigned access digits.
- Successfully complete the ground puzzle challenges to obtain the remaining access digits.
- Correctly combine the four-digit access code.
- Unlock the Intelligence Cache.
- Retrieve the Intelligence Package.
- Return to the designated Finish Zone within the prescribed mission time while complying with all safety requirements.

Teams that do not successfully complete all mandatory mission objectives shall remain eligible for ranking based on their cumulative competition score. In such cases, awards and rankings shall be determined according to the overall points obtained from the Technical Design Report, Technical Presentation, Workshop Participation, Faculty Advisor Participation, and the completed portions of the Dynamic Mission.

The Organizing Committee reserves the right to determine the final rankings in accordance with the official scoring criteria and competition regulations.

SECTION 6

TECHNICAL PRESENTATION

Like all professionals, engineers must possess a well-developed ability to synthesize issue communicate effectively to diverse audiences. The technical portion of the aero-design competit designed to emphasize the value of an ability to deliver clear, concise and effective oral presenta Teams can obtain a maximum technical presentation score of fifty (100) points. Presentation score sh comprised of scores from the presenter's delivery technique and the judges' evaluation of technical co empirical analysis, and quality visual aide.

6.1 TECHNICAL PRESENTATION REQUIREMENTS

1. Technical presentation shall last seven (7) minutes and followed by a five (5) minute "Que and Answer" (Q&A) period.
2. Technical presentation shall be delivered in English.
3. Technical presentation shall address, but are not limited to, trade studies performed, de challenges, and manufacturing techniques.
4. Technical presentation is limited to student team members only. Non-team member pilots faculty advisors can attend the technical presentation but are prohibited from participating in setup, delivery, and/or the Q&A.
5. Assistance in the use of visual aids is advisable; Film clips, if used, may not exceed one-minute duration; Film clips may not be accompanied by recorded narration.

6.2 TECHNICAL PRESENTATION PROCESS AND PROCEDURES

Each presentation room shall have a lead judge with the responsibility to ensure compliance competition rules and schedule. Lead judge will identify a timekeeper.

- With agreement from the speaker, the timekeeper will give the speaker a one
- Minute warning prior to the seven (7) minute limit.
- If the team exceeds the seven (7) minute limit, the team will be penalized.
- The presentation shall be stopped at the eleven (11) minute mark.
- A team shall have five (5) minutes for Q&A immediately following the presentation.

Questions may be asked by any judge on the panel

- Any time remaining or exceeding the 7 minutes shall be added to or subtracted from five (5) minute Q&A.
- Presentation Time Breakdown
- 3 Minutes Setup presentation, visual aide, and/or static display
- 7 Minutes Perform Technical Presentation
- 5 Minutes Questions & Answers

6.3 TECHNICAL INSPECTION

Technical and Safety inspection of all aircraft will be conducted using the published Technical and Safety Inspection checklists for each class for the current year. **The checklists can be found on the SAEISS Aero Design home page under Downloads page in the website.**

Technical and Safety Inspection is the process of checking all aircraft for:

- Compliance with all General requirements.
- Compliance with all air vehicle configuration requirements for their class.
- Overall safety and airworthiness.
- All aircraft must pass the Technical and Safety Inspection in order to compete.

It is strongly recommended to check the inspection checklist before arriving at the competition.

Evaluation Criteria	Points
Technical Knowledge	20
Design Justification	20
Mission Strategy	20
Innovation	20
Question & Answer Session	20
Total	**100**

6.4 AIRCRAFT CONFORMANCE TO 2D DRAWING

During Technical Inspection, the aircraft will be inspected and measured for conformanc the 2D drawing presented in the Design Report.

- At a minimum, dimensions will be measured and compared to the 2D drawing.
- All teams must have a hard copy of their design report with them during techn inspection.
- Models will have their max takeoff weight presented in the design report.

6.5 AIRCRAFT MUST MEET ALL INSPECTION REQUIREMENTS THROUGHOUT THE COMPETITION.

- All aircraft must meet all Technical and Safety Inspection requirements throughout competition.
- Any official may request that a drone be re-inspected if a safety requirement problem seen on the drone at any time during the event.
  - This includes any errors or omissions made by officials during inspection.

6.6 TECHNICAL AND SAFETY INSPECTION PENALTIES

No points are available to be scored as a result of the Technical and Safety

Inspection: teams may only lose points as a result of errors and problems encountered during th inspection process. Any penalties assessed during Technical Inspection will be applied to the overa competition score.